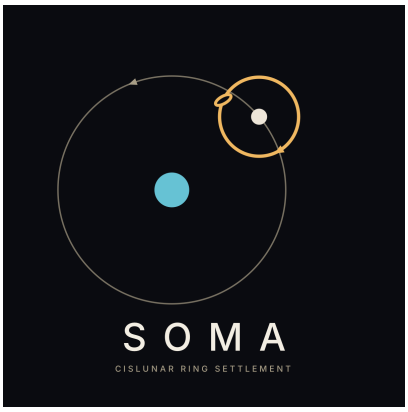


Soma: CisLunar Rings

A cislunar ring settlement design study

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3 November 2026

1 Summary

Soma is a speculative design study for a cislunar ring settlement: six rings, each 109 km around (34.7 km across), on one shared spine, housing about 3.3 million people, parked in a stable loop around the Moon. Each ring is built mostly from lunar material by self-assembling bay factories, at an estimated $87k–661k of build cost per resident. All figures are order-of-magnitude estimates for a concept, not engineering results.

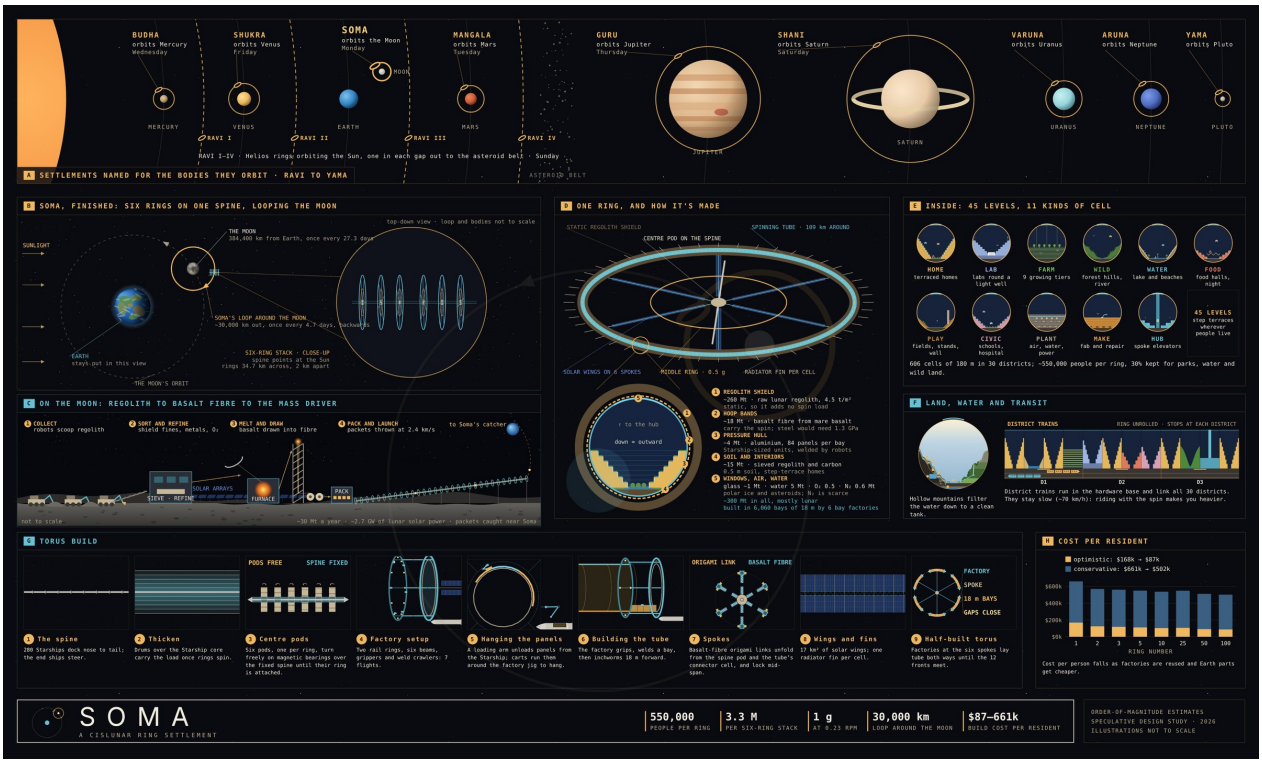


Figure 1: The Soma poster (60 × 36 in). An animated version, in which every tile plays its build step, is in the repository.

2 The design

Each ring is a 150 m-wide tube bent into a 17.35 km-radius torus that spins at 0.23 rpm for 1 g at the floor:

$$a = \omega^2 R, \qquad \omega = \sqrt{\frac{9.81\ \mathrm{m\,s^{-2}}}{17\,350\ \mathrm{m}}} \approx 0.0238\ \mathrm{rad\,s^{-1}} \approx 0.23\ \mathrm{rpm}, \qquad v = \omega R \approx 412\ \mathrm{m\,s^{-1}}. \quad (1)$$

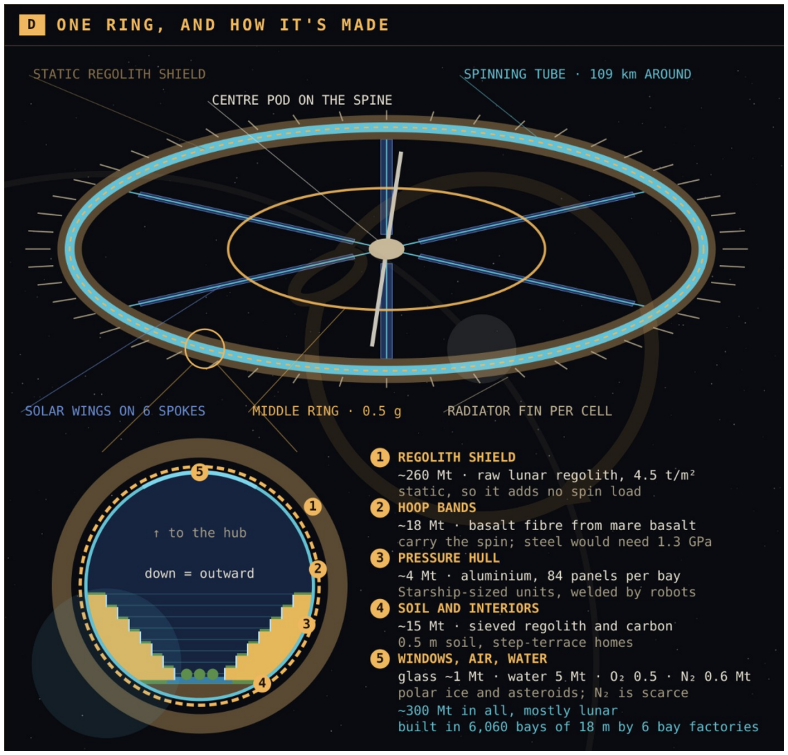


Figure 2: One ring and how it is made: shield, hoop bands, hull, soil and terraced interiors.

- **Tube and shield.** Basalt-fibre hoop bands carry the spin. The hoop stress from a ring's own mass is $\sigma = \rho v^2$, so steel would need 1.3 GPa and is too heavy. A static shell of raw regolith (∼260 Mt) shields against radiation and adds no spin load.
- **Cells.** The tube is split into 606 cells of 180 m, grouped into 30 districts. Each cell has 45 levels of 3.33 m and is one of 11 types (home, lab, farm, wild, water, food and night, play, civic, plant, make, hub).
- **Step terraces.** Every cell where people live or work steps back from a central canyon, so each level has open sky and its own terrace.
- **Spine.** 280 Starships dock nose to tail into a 14 km core, later wrapped in five concentric drums out to $r = 100$ m (∼1.4 Mt).
- **Centre pods.** Six pods, one per ring, turn on magnetic bearings around the fixed spine.
- **Spokes.** Origami-style basalt-fibre links unfold from both the pod and the tube's connector cell and lock mid-span. Spin pulls $\sigma = \rho v^2/2 \approx 0.23$ GPa on a spoke; basalt fibre holds ∼1.5 GPa.
- **Stack.** Six rings in three counter-rotating pairs, 2 km apart, so the stack has no net spin and can be steered like a gyroscope to face the Sun.
- **Power and heat.** Solar wings on the six spokes (∼6 GW) face the Sun at all times because the spin axis points at it. About 650 radiator fins, one per cell, shed heat edge-on.

- **Transit.** District trains run through the hardware base at ~70 km/h. They stay slow because riding with the spin makes you feel heavier.

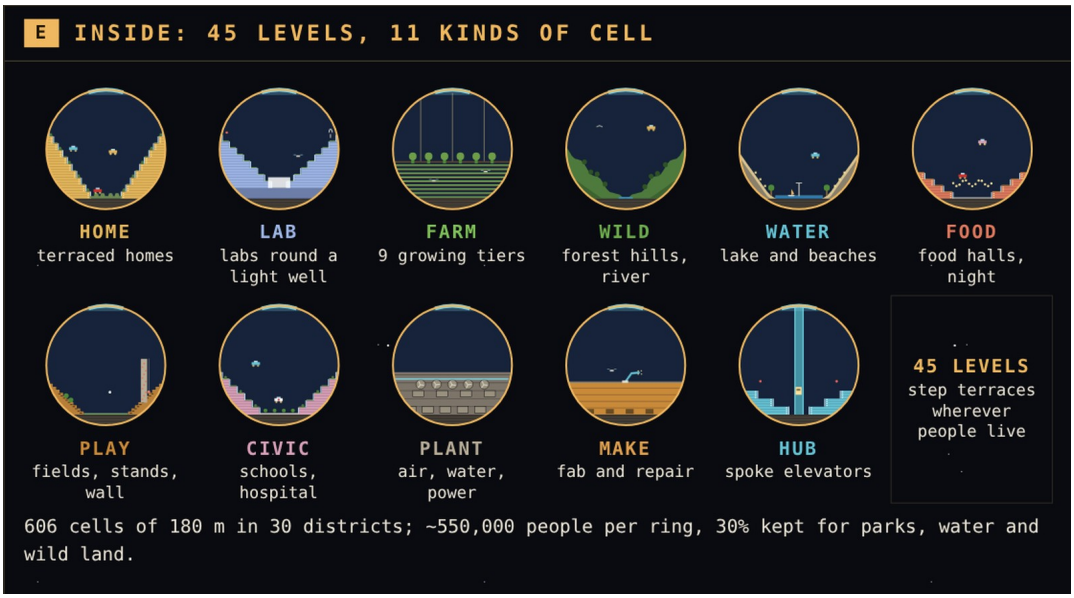


Figure 3: The 11 cell types in cross-section, with step terraces wherever people live.



Figure 4: Land, water and transit: a filter mountain and district trains in the hardware base.

3 How it is built

The build runs in nine steps, starting from the spine and ending when the tube closes into a ring. The full construction film, *How Soma is built* (5 min 57 s), is in the repository under `videos/soma-construction-film.mp4`.

1. **The spine.** Cargo Starships unload, then dock nose to tail along the spin axis; the ships at each end keep their engines to steer.
2. **Thicken.** Concentric drums are wrapped over the Starship core so it can carry Sun-tracking loads once the rings spin.
3. **Centre pods.** Six pods are built over the fixed spine one by one. They turn freely on magnetic bearings; real spin waits until the ring is attached.
4. **Factory setup.** Seven Starship flights bring two rail rings, six beams, grippers and weld crawlers for each bay factory.

5. **Hanging the panels.** A loading arm unloads hull panels from a Starship; carts run them around the factory jig to hang.
6. **Building the tube.** The factory grips the tube, welds and leak-tests one 18 m bay, then inchworms forward.
7. **Spokes.** Origami links unfold from the pod and the tube's connector cell and lock mid-span, in balanced pairs.
8. **Wings and fins.** Wing robots unroll 17 km^2 of lunar-made solar cells along the spokes; carriers fit one radiator fin per cell.
9. **Half-built torus to full ring.** Six factories build tube both ways from the spokes; the 12 fronts meet after 6,060 bays and the ring closes.

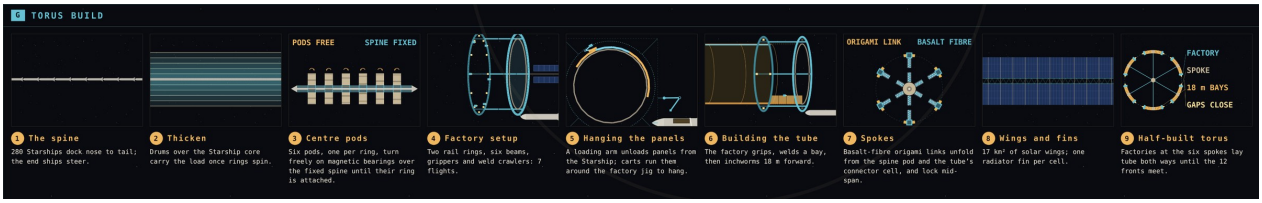


Figure 5: The nine build steps, from the spine to the half-built torus.

On the Moon, robots collect regolith, plants sort it and draw basalt into fibre, and a mass driver throws packets at 2.4 km/s to a catcher near Soma (~30 Mt a year).

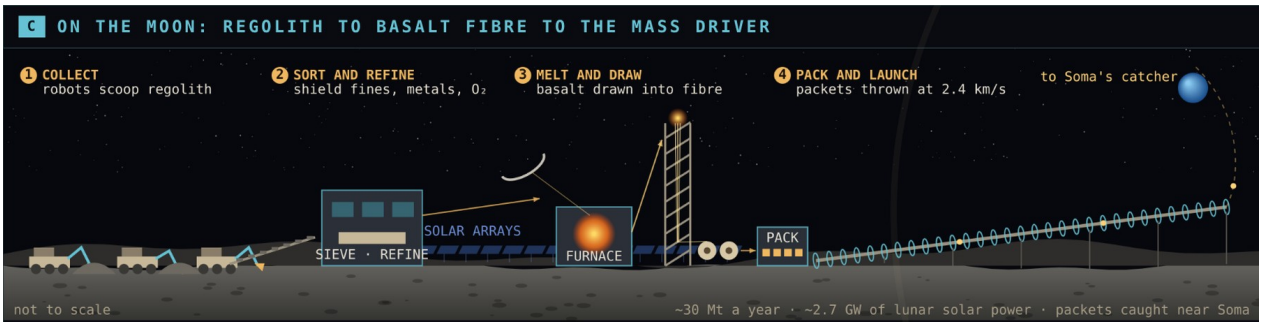


Figure 6: On the Moon: regolith to basalt fibre to the mass driver.

4 Orbit

Soma sits in a lunar distant retrograde orbit about 30,000 km from the Moon, looping it backwards every 4.7 days while the Moon carries it around Earth. In a simplified Earth–Moon–Sun model it stayed bound for 20 years with no corrections, even with a 2 m/s kick each time a ring was added. Larger loops (40,000–70,000 km) drifted off within 2–4 years. The Soma logo is drawn from this orbit.

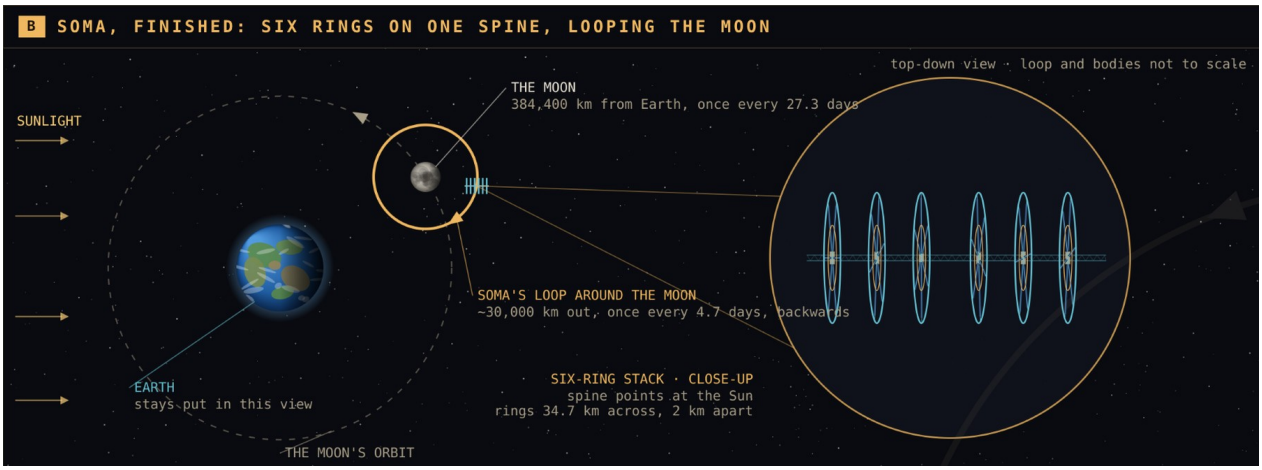


Figure 7: Soma's loop around the Moon, top-down with Earth fixed.

5 Cost per resident

Build cost per resident falls from \$168k–661k for ring 1 to \$87k–502k for ring 100, because later rings reuse the lunar industry and factories and Earth-made parts get cheaper. The range runs from an optimistic case (80% learning curve) to a conservative one (90%). A standalone ring costs about \$89–356B; 100 rings total about \$5.2T (optimistic) to \$28.7T (conservative).

6 Key numbers

Item	Value
Ring	109 km around, 17.35 km radius, 1 g at 0.23 rpm
Tube	150 m across, 45 levels of 3.33 m, 606 cells of 180 m
Population	~550,000 per ring; ~3.3 million per six-ring stack
Atmosphere	57 kPa, 34% O_2
Mass per ring	~300 Mt, mostly lunar regolith
Spine	280 Starships (14 km core) + 5 drums to $r = 100$ m, ~1.4 Mt
Middle ring	$r = 8.7$ km, 0.5 g, Ø 40 m
Stack	6 rings, 3 counter-rotating pairs, 2 km apart
Power	~6 GW per ring from spoke wings
Orbit	Lunar DRO, ~30,000 km, 4.7-day loop
Cost per resident	Ring 1: \$168k–661k → ring 100: \$87k–502k

7 Deliverables

- **Repository:** github.com/varunpra97/SomaCislunarSettlement, with the README, videos, design boards, logo and render scripts.
- **Full report:** Soma: Cost, Deployment and Operations, 36 pages.
- **Construction film:** *How Soma is built*, 5 min 57 s, from the spine to the zoom out to the orbit.
- **Poster:** 60 × 36 in print PDF, plus a 4096 px image for X.
- **Animated poster:** 8-second looping GIF in which every tile plays its build step, at 3840 px and 1920 px.

8 Caveats and open questions

These are concept estimates; the items below would need real engineering before any of the numbers hold.

- [ ] Basalt-fibre strength and creep at 1.5 GPa over decades of spin
- [ ] How the origami spoke links fold, lock and carry load at 17 km length
- [ ] Long-term orbit stability with a full-physics model, not the planar bicircular one used here
- [ ] Nitrogen supply: lunar N_2 is scarce, so it must come from asteroids or Earth
- [ ] Mass-driver catch accuracy and throughput at ∼30 Mt a year
- [ ] Cost model assumptions: learning rates, Starship launch prices and lunar industry ramp